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import tkinter as tk
from tkinter import ttk, messagebox

class GradingProgram:
    def __init__(self, root):
        self.root = root
        self.root.title("Grading Program")

        self.main_frame = tk.Frame(self.root)
        self.main_frame.pack(padx=10, pady=10)

        self.input_frame = tk.Frame(self.main_frame)
        self.input_frame.pack(fill="x")

        tk.Label(self.input_frame, text="Prelim:").grid(row=0, column=0, padx=5,
pady=5, sticky="w")
        tk.Label(self.input_frame, text="Midterm:").grid(row=1, column=0,
padx=5, pady=5, sticky="w")
        tk.Label(self.input_frame, text="Pre-Finals:").grid(row=2, column=0,
padx=5, pady=5, sticky="w")
        tk.Label(self.input_frame, text="Final Grade:").grid(row=3, column=0,
padx=5, pady=5, sticky="w")

        self.prelim_entry = tk.Entry(self.input_frame)
        self.prelim_entry.grid(row=0, column=1, padx=5, pady=5)

        self.midterm_entry = tk.Entry(self.input_frame)
        self.midterm_entry.grid(row=1, column=1, padx=5, pady=5)

        self.pre_finals_entry = tk.Entry(self.input_frame)
        self.pre_finals_entry.grid(row=2, column=1, padx=5, pady=5)

        self.final_grade_entry = tk.Entry(self.input_frame)
        self.final_grade_entry.grid(row=3, column=1, padx=5, pady=5)

        self.grade_type_frame = tk.Frame(self.main_frame)
        self.grade_type_frame.pack(fill="x")

        tk.Label(self.grade_type_frame, text="Grade Type:").pack(side="left",
padx=5)

        self.grade_type = tk.StringVar(value="Letter Grade")
        self.grade_type_combo = ttk.Combobox(
            self.grade_type_frame,
            textvariable=self.grade_type,
            values=("Letter Grade", "Numeric Grade"),
            state="readonly"
        )
        self.grade_type_combo.pack(side="left", padx=5)

        self.button_frame = tk.Frame(self.main_frame)
        self.button_frame.pack(fill="x")

        self.compute_button = tk.Button(self.button_frame, text="Compute",
command=self.compute_grade)
        self.compute_button.pack(side="left", padx=5, pady=5)
        self.reset_button = tk.Button(self.button_frame, text="Reset",
command=self.reset_fields)
        self.reset_button.pack(side="left", padx=5, pady=5)

        self.about_button = tk.Button(self.button_frame, text="About",

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command=self.about)
self.about button.pack(side="left", padx=5, pady=5)

self.output frame = tk.Frame(self.main frame)
self.output frame.pack(fill="both", expand=True)

self.output text = tk.Text(self.output frame, height=12, width=45)
self.output text.pack(padx=5, pady=5)

def compute_grade(self):
    try:
        prelim = float(self.prelim_entry.get())
        midterm = float(self.midterm_entry.get())
        pre_finals = float(self.pre_finals_entry.get())
        final grade = float(self.final grade entry.get())

        gpa = (prelim + midterm + (pre_finals * 0.5) + (final grade * 0.5))

        if gpa >= 96:
            letter_grade = "A+"
        elif gpa >= 93:
            letter_grade = "A"
        elif gpa >= 88:
            letter_grade = "B"
        elif gpa >= 83:
            letter_grade = "B-"
        elif gpa >= 78:
            letter_grade = "C"
        elif gpa >= 76:
            letter_grade = "D"
        elif gpa >= 75:
            letter_grade = "E"
        else:
            letter grade = "F"

        if gpa >= 97:
            numeric grade = "1.00"
        elif gpa >= 94:
            numeric grade = "1.25"
        elif gpa >= 90:
            numeric grade = "1.50"
        elif gpa >= 87:
            numeric grade = "1.75"
        elif gpa >= 84:
            numeric grade = "2.00"
        elif gpa >= 81:
            numeric grade = "2.25"
        elif gpa >= 78:
            numeric grade = "2.50"
        elif gpa >= 76:
            numeric grade = "2.75"
        elif gpa >= 75:
            numeric grade = "3.00"
        else:
            numeric grade = "5.00"

        if self.grade type.get() == "Letter Grade":
            display grade = letter grade
        else:

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        display_grade = numeric_grade

        remarks = "PASSED" if gpa >= 75 else "FAILED"

        self.output_text.delete(1.0, "end")
        self.output_text.insert("end", "Transaction Summary\n\n")
        self.output_text.insert("end", f"Prelim Grade : {prelim:.2f}\n")
        self.output_text.insert("end", f"Midterm Grade : {midterm:.2f}\n")
        self.output_text.insert("end", f"Pre-Finals :
{pre_finals:.2f}\n")
        self.output_text.insert("end", f"Final Grade :
{final_grade:.2f}\n")
        self.output_text.insert("end", f"GPA : {gpa:.2f}\n")
        self.output_text.insert("end", f"Displayed Grade:
{display_grade}\n")
        self.output_text.insert("end", f"Remarks : {remarks}\n")

    except ValueError:
        messagebox.showerror("Error", "Please enter numbers only.")

    def reset_fields(self):
        self.prelim_entry.delete(0, "end")
        self.midterm_entry.delete(0, "end")
        self.pre_finals_entry.delete(0, "end")
        self.final_grade_entry.delete(0, "end")
        self.output_text.delete(1.0, "end")
        self.grade_type.set("Letter Grade")

    def about(self):
        messagebox.showinfo("About", "Hello I'm Jean Claude")

if __name__ == "__main__":
    root = tk.Tk()
    app = GradingProgram(root)
    root.mainloop()

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Grading Program



Prelim:

85

Midterm:

84

Pre-Finals:

85

Final Grade:

80

Grade Type:

Letter Grade



Compute

Reset

About

Transaction Summary

Prelim Grade : 85.00
Midterm Grade : 84.00
Pre-Finals : 85.00
Final Grade : 80.00
GPA : 83.83
Displayed Grade: B-
Remarks : PASSED